

ECON 0100 | Vignette B1 | Demand

I Solutions

Answers and work are shown in the red blocks.

Q1 | Individual Demand Simulation

For each price offered to the volunteer in recitation, plot your **Quantity Demanded**, how much you would be willing to buy at that price.

There is no single right answer here: whatever the volunteer said is the data. The teaching team's script is in `Vignette_B1_simulation.md`.

Bring a crate of chocolate bars. One student comes to the front and is asked how many bars they would buy at smaller and smaller prices: at \$4 expect zero, at \$2 expect one, at \$1 expect two (one for now, one for tomorrow), and as the price drops through \$0.50, \$0.25, and \$0.10 expect many more than they could eat, some for friends and some for later. The room plots each answer as a point, price on the vertical and bars on the horizontal.

Each point is an **individual quantity demanded**, the amount one buyer is willing and able to buy at a given price. Together they trace the **individual demand curve**, or simply demand. It slopes down: the buyer's quantity demanded falls as the price rises, which is the **law of demand**. If a student notices they would have paid more than the price asked, that is consumer surplus, which waits for Block B2.

Q2 | Pumpkin Pasties

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 12 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in thousands of pasties. Plot this demand curve.

Hint: Start by finding the vertical intercept: plug in a value of 0 for quantity. Then find the horizontal intercept: plug in a value of 0 for price.

Intercepts at 12 galleons and 24 pasties.

Vertical intercept, quantity 0:

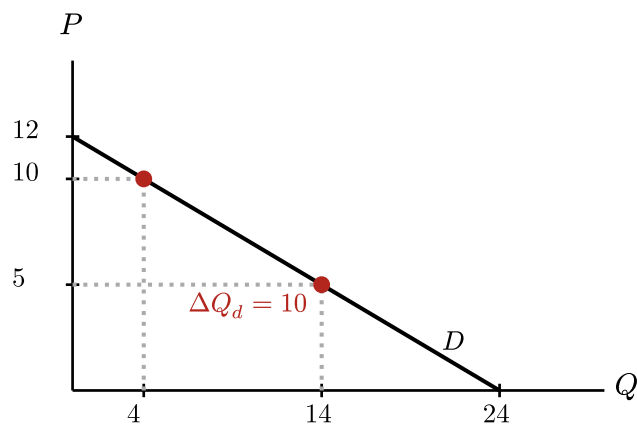
$$P_d = 12 - \frac{1}{2} \cdot 0 = 12$$

Horizontal intercept, price 0:

$$0 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 12 \rightarrow Q_d = 24$$

Demand is the straight line between (0, 12) and (24, 0): it drops by 1 galleon for every 2 pasties.

The Q3 points are marked on it.



Q3 | Quantity Demanded

a) Plot and find the quantity demanded at 10 galleons. _____

4 pasties.

Price to quantity: plug the price into the demand curve and solve for Q_d .

$$10 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 2 \rightarrow Q_d = 4$$

b) Plot and find the quantity demanded at 5 galleons. _____

14 pasties.

$$5 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 7 \rightarrow Q_d = 14$$

c) How much did quantity demanded change as the price dropped from 10 to 5? _____

Up by 10 pasties.

$\Delta Q_d = 14 - 4 = 10$: the price fell and quantity demanded rose, a movement down along the demand curve from (4, 10) to (14, 5). That is the law of demand from Q1 again, now for the whole market rather than one buyer.